

CertLI and EigenLI: nDCG@10

Follow-up to *CertLI: First Experiments*

Complete code repository: github.com/RS-010806/certLI

Short answer: yes, in two ways.

1. **Same memory.** Shifting by the corpus mean improves EigenLI on AnswerAI-small by 2.2 nDCG@10 points on SciFact and 3.8 on NFCorpus. On ColBERTv2 it does not help (Table 1).
2. **With exact rescoring.** Under the same guarantee, CertLI reaches exact MaxSim quality while rescoring up to 10 times fewer documents than an EigenLI first stage, at about 2.5 times EigenLI’s memory (Table 2).

Setup

EigenLI as defined in its paper [1]: each document keeps the top- k eigenvectors of its token second-moment matrix $\sum_j d_j d_j^\top$, and $s(Q, D) = \sum_i \|\Pi_D q_i\|^2$ (my re-implementation). Same models, corpora and protocol as the main report. At $k = 32$, EigenLI stores 13.6% of the full fp16 index.

1. Same memory, no rescoring

Both of our variants keep EigenLI’s budget of 32 vectors per document. *Mean-shifted EigenLI* subtracts one corpus-wide mean from documents and queries before building the subspace and scoring (CertLI’s Lemma 1 shift, not per-text centring). *Low-rank + sparse EigenLI* keeps 24 eigenvectors plus the 8 tokens farthest from that subspace, scored as $\sum_i \max(\|\Pi_D q_i\|^2, \max_t \langle q_i, t \rangle_+^2)$.

Table 1: nDCG@10 at equal memory (32 vectors per document). Bold: best of the three methods; in brackets, change against EigenLI.

Model	Data	EigenLI	Mean-shifted	Low-rank + sparse	Exact
			EigenLI (ours)	EigenLI (ours)	MaxSim
ColBERTv2	SciFact	67.7	66.6 (−1.1)	67.0 (−0.7)	69.3
ColBERTv2	NFCorpus	31.9	31.0 (−0.9)	32.0 (+0.1)	34.4
AnswerAI-small	SciFact	67.4	69.6 (+2.2)	68.9 (+1.5)	74.5
AnswerAI-small	NFCorpus	28.0	31.8 (+3.8)	27.5 (−0.5)	37.0

The mean shift helps AnswerAI-small, whose tokens share one dominant direction, and closes 31% (SciFact) and 42% (NFCorpus) of its gap to exact MaxSim. On ColBERTv2, neither variant helps meaningfully (−1.1 to +0.1).

2. With exact rescoring

Both pipelines rescore with exact MaxSim until Learn-then-Test [2] certifies that at most 5% of queries get a top-10 different from exact MaxSim (90% confidence). Only the first stage differs: EigenLI (13.6% of the index) or the CertLI index with its bounds (34% to 37%).

Table 2: Calibrated exact rescoring under the same guarantee. Rescored: share of documents rescored per query. Bold: better of the pair.

Model	Data	EigenLI first stage		CertLI (ours)		Exact MaxSim
		nDCG@10	Rescored	nDCG@10	Rescored	
ColBERTv2	SciFact	69.1	8.7%	69.3	1.4% (6.2× fewer)	69.3
ColBERTv2	NFCorpus	34.1	73.1%	34.0	7.1% (10.3× fewer)	34.4
AnswerAI-small	SciFact	73.4	24.4%	74.7	12.6% (1.9× fewer)	74.5
AnswerAI-small	NFCorpus	36.6	78.1%	36.6	71.0% (1.1× fewer)	37.0

Both reach exact MaxSim quality within 0.4 points, but CertLI rescores 1.1 to 10.3 times fewer documents. Without rescoring, the CertLI index alone also beats EigenLI in all four settings (by 1.1 to 3.3 points), at about 2.5 times the memory.

Takeaway. EigenLI is the stronger compact scorer, and the mean shift is a free gain for anisotropic models such as AnswerAI-small. CertLI is the cheaper route to exact MaxSim quality when more memory is available.

References

- [1] Archish S, S. Basu, A. Garg, R. Krishnaswamy, K. Shiragur. EigenLI: spectral approximations to late interaction. arXiv:2609.07561, 2026. [2] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei. Learn then Test: calibrating predictive algorithms to achieve risk control. arXiv:2110.01052, 2021.